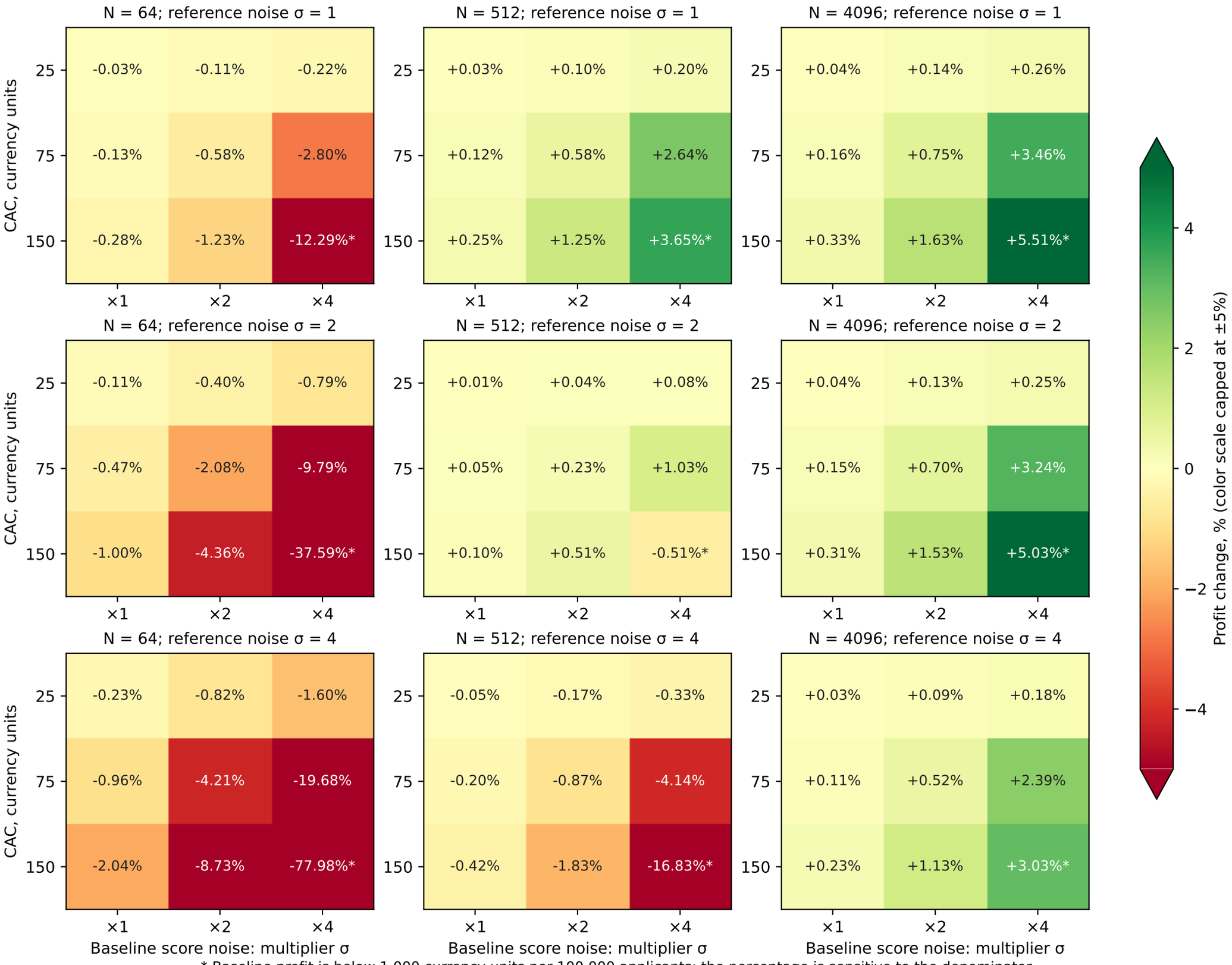
When does an additional score reduce profit by more than 1%?
Change in mean profit relative to baseline; all 81 scenarios
N = 64; reference noise σ = 1
N = 512; reference noise σ = 1
N = 4096; reference noise σ = 1
-0.03%
-0.11%
-0.22%
-0.13%
-0.58%
-2.80%
-0.28%
-1.23%
-12.29%*
+0.03%
+0.10%
+0.20%
+0.12%
+0.58%
+2.64%
+0.25%
+1.25%
+3.65%*
+0.04%
+0.14%
+0.26%
+0.16%
+0.75%
+3.46%
+0.33%
+1.63%
+5.51%*
N = 64; reference noise σ = 2
N = 512; reference noise σ = 2
N = 4096; reference noise σ = 2
-0.11%
-0.40%
-0.79%
-0.47%
-2.08%
-9.79%
-1.00%
-4.36%
-37.59%*
+0.01%
+0.04%
+0.08%
+0.05%
+0.23%
+1.03%
+0.10%
+0.51%
-0.51%*
+0.04%
+0.13%
+0.25%
+0.15%
+0.70%
+3.24%
+0.31%
+1.53%
+5.03%*
N = 64; reference noise σ = 4
N = 512; reference noise σ = 4
N = 4096; reference noise σ = 4
-0.23%
-0.82%
-1.60%
-0.96%
-4.21%
-19.68%
-2.04%
-8.73%
-77.98%*
-0.05%
-0.17%
-0.33%
-0.20%
-0.87%
-4.14%
-0.42%
-1.83%
-16.83%*
+0.03%
+0.09%
+0.18%
+0.11%
+0.52%
+2.39%
+0.23%
+1.13%
+3.03%*
CAC, currency units
25
75
150
×1
×2
×4
Baseline score noise: multiplier σ
Profit change, % (color scale capped at ±5%)
4
2
0
−2
−4
* Baseline profit is below 1,000 currency units per 100,000 applicants; the percentage is sensitive to the denominator.